

Achivement Test 2: Control Systems I

Duration: 90 minutes

Answer all questions. Show essential steps.

Question 1. Modeling of a Permanent-Magnet DC Motor (30 points)

Consider a permanent-magnet DC motor. The magnetic field is generated by permanent magnets, and no field current exists.

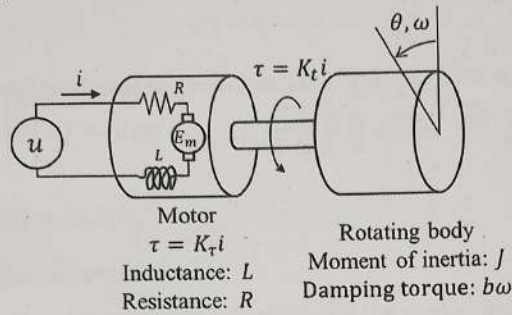


Figure 1 A motor driving a rotating body and its equivalent armature circuit

Figure 1 shows a rotating body driven by the motor with the equivalent armature circuit. The armature circuit consists of an inductance L , a resistance R , and a back electromotive force $e = K_e \omega$ opposing the applied voltage u . It generates torque τ proportional to the electric current i . The rotor has inertia J and is subject to viscous damping torque $b\omega$ in the direction opposite to the rotation.

The electrical dynamics of the armature circuit are given by

$$L \frac{di}{dt} + R_i + K_e \omega = u$$

Rotational dynamics are governed by

$$J \frac{d\omega}{dt} + b\omega = K_t i$$

The angular position θ satisfies

$$\frac{d\theta}{dt} = \omega$$

Define the state vector $\mathbf{x} = [\theta, \omega, i]^T$. Assume that the angular position θ and the armature current i are measured, and the angular velocity ω is not directly measured. Thus, the output vector is defined as $\mathbf{y} = [\theta, i]^T$.

Derive the state-space model $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ and the output equation $\mathbf{y} = \mathbf{C}\mathbf{x}$. Explicitly give the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ using the symbols.

Question 2. Observability, State Feedback, and Observer Design (40 points)

Consider the following state-space model, which represents a physical system of the same structure as the motor model in Question 1, but with numerical parameters already substituted.

$$\dot{x} = Ax + Bu$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C A =$$

$$y = Cx$$

The state vector is $x = [x_1, x_2, x_3]^T$. The following sensor candidates are available:

$$S1: y = x_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$S2: y = x_2$$

$$S3: y = x_3$$

- (1) (20 points) For each sensor S1–S3, write the corresponding output matrix C , construct the observability matrix $W_O = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$, and compute its rank. Determine which sensor makes the system observable.

- (2) (10 points) Consider the state feedback $u = -Kx$ with $K = [0, 2, 1]$. Compute the eigenvalues of the closed-loop matrix $A - BK$. Based on the eigenvalues obtained, comment briefly on whether this feedback is suitable for stabilization.

- (3) (10 points) Using the output $y = Cx$ that you identified as observable in part (1), design a full-order observer

$$\dot{z} = Az + Bu + L(y - Cz)$$

$$A - K_0 C$$

$$A - K_0 C$$

such that the eigenvalues of the observer error dynamics are $\{-4, -5, -6\}$. You may use the observable canonical form provided in the reference sheet.

$$A - BK$$

λ^2	9λ	20
λ	23	20λ
$+6$	$6\lambda^2$	54λ
		120

$$\begin{vmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & -3 & \lambda+2 \end{vmatrix}$$

$$\begin{vmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & -3 & \lambda+2 \end{vmatrix}$$

$$\begin{vmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & -3 & \lambda+2 \end{vmatrix}$$

Question 3. Optimal Control and Interpretation of Responses (20 points)

Consider the linear system

$$\dot{x} = Ax + Bu$$

An infinite-horizon linear quadratic regulator (LQR) is used with the cost function

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, Q = \text{diag}(q_1, q_2, q_3) \text{ and } R = r > 0.$$

The initial condition is $x(0) = [0, 1, 0]^T$.

Three candidate weight matrices are given:

W1: $Q = \text{diag}(1, 1, 1), R = 1$

W2: $Q = \text{diag}(1, 100, 1), R = 1$

W3: $Q = \text{diag}(1, 1, 1), R = 100$

Four response plots (A–D) are provided separately on next two pages. Each plot set shows the response of $x_1(t)$, $x_2(t)$, $x_3(t)$ and the control input $u(t)$.

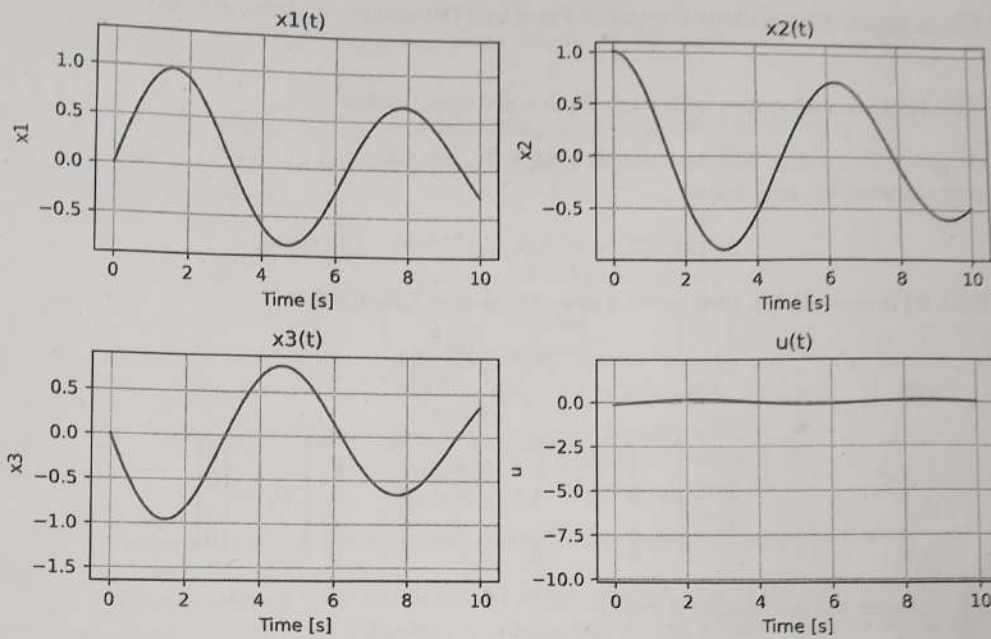
(1) (15 points) For each response plot A–D, select all weight matrices (W1–W3) that could have produced the response. If none apply, clearly indicate so.

(2) (5 points) One of the response plots cannot be generated by any of the above weight matrices. Identify this plot and briefly justify your answer in terms of the trade-off between state regulation and control effort.

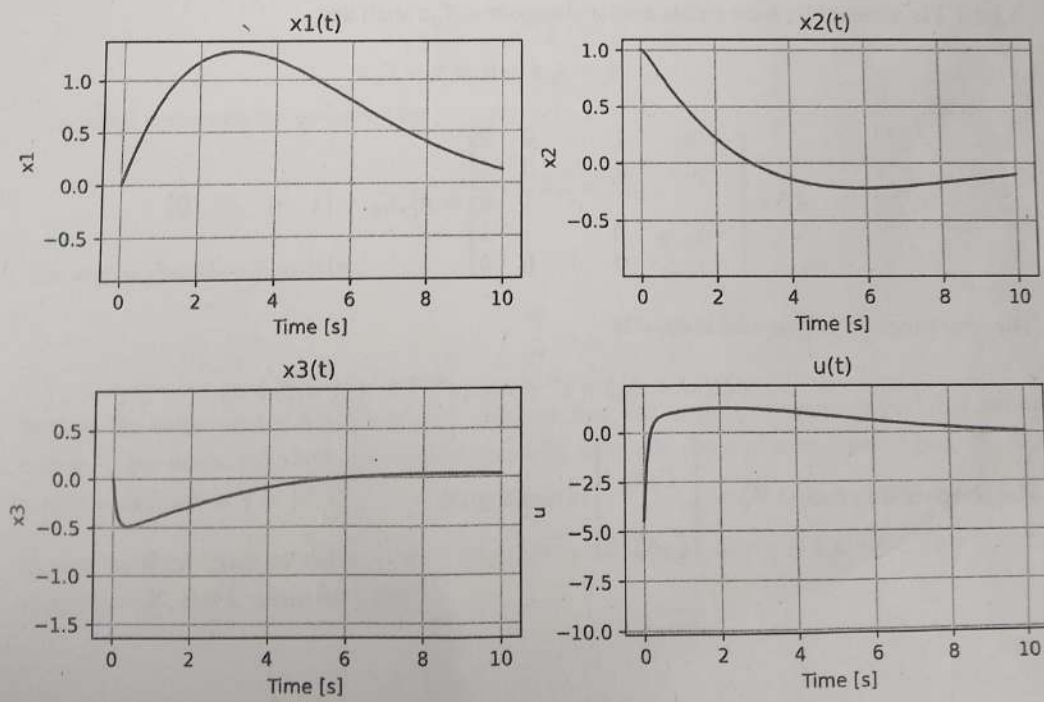
$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -1 \end{bmatrix} x, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$Q =$$

Response C



Response D



Cheat Sheet: Controllable Canonical Form and Observable Canonical Form

For a linear control system with single input u and single output y ,

$$\dot{x} = Ax + Bu, y = Cx,$$

with characteristic polynomial

$$\det(sI - A) = s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0,$$

If (A, B) is controllable, there exists a state change $x = T_c z$ such that

$$\dot{z} = A_c z + B_c u$$

with

$$A_c = \begin{bmatrix} -a_{n-1} & -a_{n-2} & \dots & -a_1 & -a_0 \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix}, B_c = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The characteristic polynomial is exactly

$$\det(sI - A_c) = s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0$$

The controllability matrix $W_c = [B_c, A_c B_c, \dots, A_c^{n-1} B_c]$ is nonsingular.

If (A, C) is observable, there exists a state change $x = T_o z$ such that

$$\dot{z} = A_o z + B_o u, y = C_o z$$

with

$$A_o = \begin{bmatrix} -a_0 & 1 & \dots & 0 & 0 \\ -a_1 & 0 & \ddots & 0 & \vdots \\ \vdots & & \ddots & 1 & 0 \\ -a_{n-2} & 0 & & \ddots & 1 \\ -a_{n-1} & 0 & \dots & 0 & 0 \end{bmatrix} = A_c^T, C_o = [1 \quad 0 \quad \dots \quad 0]$$

The characteristic polynomial is exactly

$$\det(sI - A_o) = s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0$$

The observability matrix $W_o = \begin{bmatrix} C_o \\ C_o A_o \\ \vdots \\ C_o A_o^{n-1} \end{bmatrix}$ is nonsingular.